SOP Number and version: SOP-NAT-TOUCANPREP.v01

SOP Title: **TOUCAN-MRSA Stock Solution Preparation**

Table 1. SOP current version review and approval table

	NAME	TITLE	SIGNATURE	DATE
Author	Baris Ata Borsa	Senior Researcher		19/02/2026
Reviewer				
Approval	Frank J. Hernandez	Principal investigator		20/02/2026

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Approved by PI (Frank J. Hernandez) - 2026-02-19T21:48:38.635Z

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1. Overview and background

TOUCAN-MRSA is the leading molecule of the TOUCAN Platform. It is the first developed targeted antibiotic that is effective in treating Staphylococcus aureus infections. TOUCAN-MRSA is an oligonucleotide prodrug and appropriate preparation of a stock solution using the lyophilized oligonucleotide product is crucial.

2. Scope

In this SOP, preparation of a TOUCAN-MRSA stock solution using a lyophilized product in Nucleic Acids Technology Laboratories located at Linköping University (Campus Valla, B-huset, Rum 2E:686) will be explained.

As an example, the order (00313890) produced on 27.05.2022 by Biomers.net will be used.

This SOP covers the sections below.

- Calculation of the concentration equal to floxuridine
- Preparation of stock solution

3. Procedure

3.1. Calculation

- Divide calculated molecular weight of TOUCAN-MRSA by the molecular weight of floxuridine (246,19 g/mol) to find **the TOUCAN coefficient.**
    - 2133 g/mol ÷ 246,19 g/mol = 8,66 (the TOUCAN coefficient)
- Divide the total weight of TOUCAN-MRSA into the coefficient found in previous step to find the **total floxuridine weight in the product.**
    - 110558,7 µg ÷ 8,66 = 12760,64 µg = 12,76 mg (**order 00313890**)
- Calculate the buffer volume that needs to be added to make a stock solution of 10 mg/ml.
    - 1,276 ml of buffer is needed to prepare a 10 mg/ml stock solution.

3.2. Drug preparation

- Use PBS+/+ as buffer and add the calculated volume in the tube with lyophilized oligonucleotide.
- Vortex well until it is homogenous.
- Wait for the entire dissolved product to go down the tube and vortex-generated bubbles to disappear or spin it down.
- Transfer the volume into another tube that is adequately labeled.
- Prepare desired aliquots or different concentrations by diluting the product using PBS+/+, if needed.
- Store all TOUCAN-MRSA solutions at -20°C until use.

4. Purpose/Objective

The objective of this SOP is to guide Nucleic Acids Technology staff on preparing a TOUCAN-MRSA stock solution in Nucleic Acids Technology Laboratories.

5. Definition / Explanation of key terms

- SOP – Standard Operating Procedure
- TOUCAN – Therapeutic Oligonucleotides Activated by Nucleases
- MRSA – Methicillin Resistant Staphylococcus aureus
- PPE – Personal Protective Equipment
- PBS – Phosphate Buffered Saline

- µg – Microgram
- mL – Milliliter
- mg – Milligram

6. Responsibilities

This SOP was prepared by Baris Ata Borsa and approved by Frank J. Hernandez, principal investigator.

7. Equipment/Material

- Gloves
- Pipettes
- Pipette tips
- Eppendorf tubes
- Lyophilized TOUCAN-MRSA
- PBS+/+
- Vortex

8. Caution/Warning

- Use PPE according to **SOP-NAT-BSL2**.
- Clean the working area before and after work.

9. References

Dissolving and Storage. Dissolving of oligonucleotides. Biomers.net webpage. https://www.biomers.net/en/Technical_Information/Dissolving_and_storage.html#:~:text=Annealing%20of%20oligonucleotides-,Dissolving%20of%20oligonucleotides,to%20depurination%20of%20the%20oligonucleotide.

10. Version Control

Original version of the **SOP-NAT-TOUCANPREP.v01**

11.Attachments/Appendices

Appendix I:

- The datasheet of TOUCAN-MRSA order 00313890 that is used as an example in this SOP.